

DOOM-SED: Downstream-Oriented Open-Vocabulary Model for Sound Event Detection

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Abstract—This paper presents DOOM-SED, a downstream-oriented open-vocabulary model for sound event detection. Open-vocabulary SED aims to detect sound events described by arbitrary textual labels. Existing methods have been formulated as query-based detection, in which each target class is specified and detected independently. This formulation, however, ignores relationships among semantically related target labels, which often increases crosstalk errors. In this paper, we propose a downstream-oriented training framework and network architecture that explicitly handle a given set of target labels. Specifically, we train a detection model to solve dynamically generated detection episodes that mimic downstream SED tasks with a small set of target classes. The model is designed and trained to capture inter-label relationships through self-attention over label embeddings. Each episode is constructed on the fly from temporally sound event labels in AudioSet Strong and carefully designed based on the hierarchical structure of the AudioSet ontology. Experimental results on the publicly available DESED, URBAN-SED, and MAESTRO Real benchmarks demonstrate that DOOM-SED significantly improves downstream performance over existing open-vocabulary models.

Index Terms—Sound event detection, open-vocabulary models

1. INTRODUCTION

Sound event detection (SED) aims to identify target sound events with their onset and offset temporal boundaries [1]–[5]. While SED has been studied for a wide range of machine listening applications [6]–[8], the set of target sound events typically varies across applications, such as surveillance, multimedia retrieval, and human-robot interaction. Such application-dependent requirements call for SED models that can detect sound events specified by users at inference time, beyond a predefined set of event classes. Open-vocabulary SED provides this flexibility by leveraging audio-language models [9], [10] that allow users to dynamically specify target sound events through textual labels [11]–[13].

Standard architectures for open-vocabulary SED are designed to fuse frame-level audio features and textual label features extracted separately [11]–[15]. For example, frame-level extensions of contrastive language-audio pretraining (CLAP) [9], [10] predict event probabilities using dot-product similarities between audio and text features [11]–[13]. FlexSED [14] improved detection performance by conditioning audio features on textual label features rather than relying only on audio-text similarity. Overall, prior work has mainly focused on modeling the correspondence between individual target labels and audio features. However, existing open-vocabulary models still exhibit

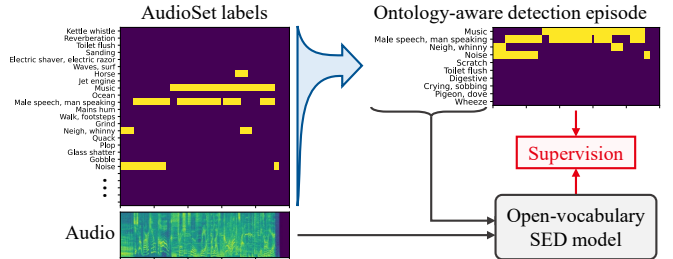


Fig. 1: Overview of our downstream-oriented training.

a performance gap on SED benchmarks compared with models trained on the corresponding benchmark data [16]–[18].

One major challenge in applying open-vocabulary SED to unseen downstream tasks is the variation in label taxonomies. Identical or semantically similar sound events may be assigned different labels across datasets. For example, walking-related sound events are labeled as “footsteps” in MAESTRO Real [18] but as “people walking” in TUT Sound Events 2016 [19]. In addition, even identically named labels may refer to sound events at different levels of granularity across datasets. For example, DESED [16] defines sounds related to human voices as a single “speech” class. In contrast, MAESTRO Real distinguishes between “children voices” and “people talking,” such that the latter should exclude children’s voices. These observations suggest that target events depend on the relationships among the labels defined within each dataset, which have received limited attention in existing studies.

In this paper, we propose DOOM-SED, a downstream-oriented open-vocabulary SED model that explicitly captures relationships among target labels. As illustrated in Fig. 1, we train the model on dynamically generated detection episodes, each consisting of an input–target pair that mimics a downstream SED task. For each episode, a small set of target labels is sampled from the AudioSet Strong label vocabulary [20] based on its ontology [21], and the corresponding frame-level targets are generated. The model is trained to capture inter-label relationships using self- and cross-attention over label and audio features. Through this downstream-oriented training, DOOM-SED learns not only label–audio correspondences but also relationships among the target labels.

The main contribution of this study is to advance open-vocabulary SED toward strong generalization to practical downstream applications. Our downstream-oriented extensions are

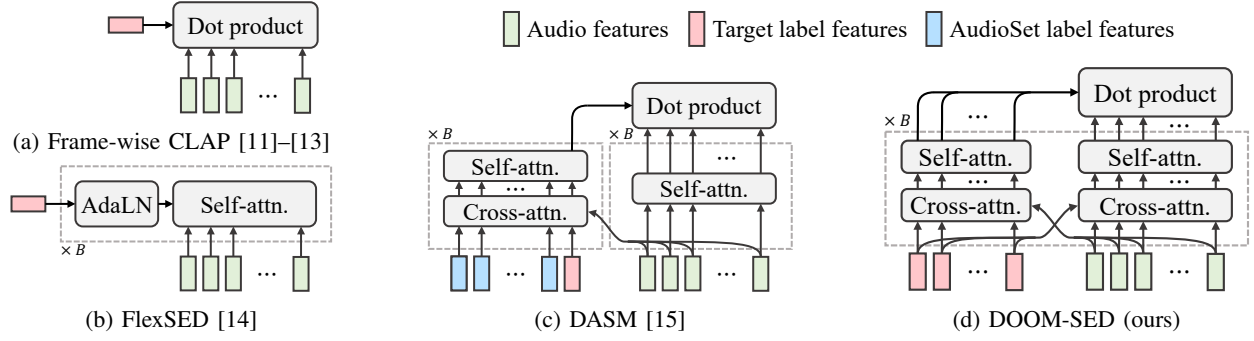


Fig. 2: Comparison of network architectures for open-vocabulary SED models. We omitted feed-forward layers for simplicity.

designed to address scenarios involving a small set of target labels through both the model architecture and training procedure. Experimental evaluations on the publicly available DESED [16], URBAN-SED [17], and MAESTRO Real [18] datasets show that DOOM-SED markedly improves the downstream detection performance of open-vocabulary SED without training on task-specific data. The model weights and source code are publicly available at <https://yband.jp/projects/doom-sed>.

2. BACKGROUND

We introduce open-vocabulary SED methods from the perspectives of network architectures and training schemes.

2.1. Network architectures for open-vocabulary SED

As illustrated in Fig. 2(a)–(c), existing open-vocabulary SED models can be categorized into three types according to how they integrate audio and textual label features [11]–[15]. Many existing models, including FLAM [11], FineLAP [12], and PE_{A-Frame} [13], utilize the frame-wise CLAP architecture shown in Fig. 2(a). FlexSED [14] is a representative model of text-conditioning architectures, as illustrated in Fig. 2(b). It uses adaptive layer normalization (AdaLN) [22] to condition audio features on textual label features. Another architecture, proposed in DASM [15] and illustrated in Fig. 2(c), combines self-attention among label features with cross-attention between label and audio features. However, at inference time, its self-attention allows target labels to interact only with labels observed during training (i.e., 407 AudioSet Strong labels) but not among themselves. In contrast to these architectures, our model aims to explicitly capture relationships among the target labels provided at inference time.

2.2. Training techniques for open-vocabulary SED

Most existing open-vocabulary SED models have been trained on AudioSet Strong [20]. This dataset provides more than 400 event labels organized in a hierarchical sound-event ontology [21]. DASM is trained to predict frame-level sound activities for all AudioSet Strong labels by using the entire label set as queries. During training, each active ground-truth label is propagated to its corresponding higher-level categories in the AudioSet ontology to preserve semantic consistency. In contrast, FlexSED [14] and FineLAP [12] are trained to capture audio–text correspondences by using a small query set

consisting of positive labels and selected negative labels. The negative labels are selected to be semantically distinct from the positive labels based on the AudioSet ontology. Our study instead uses the ontology to dynamically construct detection episodes, each comprising a small and semantically consistent set of positive and negative labels.

3. DOWNSTREAM-ORIENTED OPEN-VOCABULARY MODEL FOR SOUND EVENT DETECTION

We propose a downstream-oriented architecture and training scheme that explicitly model relationships among target labels.

3.1. Downstream-oriented network architecture

Our DOOM-SED f_θ takes frame-wise audio features $\mathbf{z}_t \in \mathbb{R}^D$ and textual label features $\mathbf{w}_k \in \mathbb{R}^D$ as input and predicts posterior probabilities of the corresponding events $\hat{y}_{kt} \in [0, 1]$:

$$\{\hat{y}_{kt}\}_{k,t=1}^{K,T} \leftarrow f_\theta(\{\mathbf{z}_t\}_{t=1}^T, \{\mathbf{w}_k\}_{k=1}^K), \quad (1)$$

where θ represents the set of model parameters, D is the feature dimension, and T and K denote the numbers of time frames and target labels, respectively. The audio and textual label features are extracted by the audio and text encoders, respectively, as described in Sec. 4.2.

As illustrated in Fig. 2(d), we explicitly model the information exchange across the audio and label features with a dual cross-attention layer (CrossAttn) [23], [24] as follows:

$$\{\mathbf{z}_t\}_{t=1}^T \leftarrow \text{CrossAttn}(\{\mathbf{z}_t\}_{t=1}^T, \{\mathbf{w}_k\}_{k=1}^K), \quad (2)$$

$$\{\mathbf{w}_k\}_{k=1}^K \leftarrow \text{CrossAttn}(\{\mathbf{w}_k\}_{k=1}^K, \{\mathbf{z}_t\}_{t=1}^T). \quad (3)$$

The latent features are then fed to modality-wise self-attention (SelfAttn) [23] to capture relationships within each modality:

$$\{\mathbf{z}_t\}_{t=1}^T \leftarrow \text{SelfAttn}(\{\mathbf{z}_t\}_{t=1}^T), \quad (4)$$

$$\{\mathbf{w}_k\}_{k=1}^K \leftarrow \text{SelfAttn}(\{\mathbf{w}_k\}_{k=1}^K). \quad (5)$$

The operations in Eqs. (2)–(5), followed by convolution-based feed-forward layers [23], [25], [26], are repeated B times. The event predictions $\hat{y}_{kt}$ are finally obtained by taking a dot product of $\mathbf{z}_t$ and $\mathbf{w}_k$ followed by the sigmoid function:

$$\hat{y}_{kt} \triangleq \text{Sigmoid}(\mathbf{z}_t^\top \cdot \mathbf{w}_k). \quad (6)$$

Algorithm 1: Ontology-Aware Episode Sampling

Input: AudioSet Strong labels $\{y_{ct}\}_{t,c=1}^{T,C}$, number of output labels K , $\mathcal{A} : c \mapsto \{\text{ancestor labels of } c\}$ and $\mathcal{D} : c \mapsto \{\text{descendant labels of } c\}$
Output: Positive labels $\mathcal{P}$ and negative labels $\mathcal{N}$

▷ Initialize positive and negative candidates
1 $\mathcal{P} \leftarrow \{c \mapsto y_{c1:T} \mid \max_t y_{ct} > 0\}$, $\mathcal{N} \leftarrow \{c \mapsto y_{c1:T} \mid \max_t y_{ct} = 0\}$;
▷ (a) Resolve positive-negative ontology conflicts
2 $\mathcal{N} \leftarrow \mathcal{N} \setminus \bigcup_{c \in \text{keys}(\mathcal{P})} (\mathcal{A}(c) \cup \mathcal{D}(c))$;
▷ (b) Resolve parent-child conflicts of positives
3 **for** $c \in \text{keys}(\mathcal{P})$:
4 **if** $c \notin \text{keys}(\mathcal{P})$:
5 **continue**;
6 $M_c \leftarrow \text{keys}(\mathcal{P}) \cap \mathcal{D}(c)$;
7 **if** $M_c \neq \emptyset$:
8 **if** $\text{rand}(0, 1) < 0.5$:
9 $\mathcal{P} \leftarrow \mathcal{P} \setminus \{c\}$;
10 **else**
11 $\mathcal{P}(c) \leftarrow \max(\{\mathcal{P}(c)\} \cup \{\mathcal{P}(m) \mid m \in M_c\})$;
12 $\mathcal{P} \leftarrow \mathcal{P} \setminus M_c$;
▷ (c) Ontology-based label replacement
13 **for** $c \in \text{keys}(\mathcal{P})$:
14 **if** $\mathcal{A}(c) \neq \emptyset$ **and** $\text{rand}(0, 1) < 0.2$:
15 Sample u uniformly from $\mathcal{A}(c)$;
16 $\mathcal{P}(u) \leftarrow \mathcal{P}(c)$;
17 $M_u \leftarrow \text{keys}(\mathcal{P}) \cap \mathcal{D}(u)$;
18 $\mathcal{P}(u) \leftarrow \max(\{\mathcal{P}(u)\} \cup \{\mathcal{P}(m) \mid m \in M_u\})$;
19 $\mathcal{P} \leftarrow \mathcal{P} \setminus M_u$;
20 $\mathcal{N} \leftarrow \mathcal{N} \setminus (\{u\} \cup \mathcal{D}(u))$;
▷ Sample an episode with K labels
21 $\mathcal{P} \leftarrow \text{UniformSample}(\mathcal{P}, \min(K, |\mathcal{P}|))$;
22 **if** $|\mathcal{N}| < K - |\mathcal{P}|$:
23 **return** RETRY
24 $\mathcal{N} \leftarrow \text{UniformSample}(\mathcal{N}, K - |\mathcal{P}|)$;
25 **return** $\mathcal{P}, \mathcal{N}$

3.2. Downstream-oriented training scheme

Our method trains the SED model on detection episodes dynamically generated to mimic the downstream tasks. As summarized in Algorithm 1, each episode is generated from AudioSet Strong, which provides temporally annotated event labels $\{y_{ct} \in \{0, 1\}\}_{t,c=1}^{T,C}$, where C is the number of event classes. It also uses set-valued mappings $\mathcal{A}$ and $\mathcal{D}$, which return the ancestor and descendant labels of class c , respectively, based on the AudioSet ontology [21]. Using the ontology to reflect the hierarchical relationships among event classes, the episode generation consists of the following four steps.

Candidate initialization: We first construct positive and negative candidate mappings $\mathcal{P}$ and $\mathcal{N}$, respectively, according to the presence or absence of each class (Line 1).

Ontology-conflict resolution: We then remove negative candidates that are ancestors or descendants of any positive label from $\mathcal{N}$ to resolve positive-negative conflicts (Line 2). On the other hand, multiple positive candidates may have ancestor-descendant relationships. We either merge the descendants into their ancestor or remove the ancestor while retaining its descendants, with equal probability (Lines 3–12).

Ontology-based label replacement: To train the model to

handle different label granularities, we randomly replace positive labels with their ancestors in the ontology (Lines 13–20). Note that any positive descendants of the selected ancestor are merged into it, while the ancestor and its descendants are removed from $\mathcal{N}$ to avoid semantic conflicts.

Episode sampling: Finally, we sample up to K positive candidates and fill the remaining slots with negative candidates to K labels (Lines 21–25). If there are insufficient negative candidates, we retry episode generation with another sample.

The resulting episodes are used to train the open-vocabulary model with binary cross-entropy loss.

4. EXPERIMENTAL EVALUATION

We evaluated the proposed method on downstream tasks using DESED [16], URBAN-SED [17], and MAESTRO Real [18].

4.1. Datasets

Our model was trained on AudioSet Strong [20], a large-scale dataset of diverse sound events. It consists of 10-second clips annotated with 456 sound event classes defined within the AudioSet ontology. We collected 100,018 clips for training and 16,400 for validation. We also optionally trained our model on the simulated audio clips used in FineLAP [12]. This dataset consists of 100,000 clips, with foreground events sourced from FSD50K [27] and background signals drawn from the Adobe Audition SFX Library [28].

We evaluated detection performance on downstream tasks using three established SED benchmarks: DESED [16], URBAN-SED [17], and MAESTRO Real [18]. DESED [16] provides 10-second audio clips recorded in domestic environments and covers 10 sound event classes. We used its real validation and public evaluation subsets, consisting of 1,168 and 692 clips, respectively. URBAN-SED [17] provides 10-second synthetic audio clips simulating urban soundscapes and covers 10 sound event classes. We used its validation and evaluation subsets, each consisting of 2,000 clips. Finally, MAESTRO Real [18] is a real-world SED dataset with crowd-sourced soft labels reflecting multiple annotators’ perceptions. Following DCASE Challenge 2024 Task 4 [29], we used its validation set of 718 10-second clips with binarized annotations for 11 sound event classes. All audio signals were resampled to 16 kHz.

4.2. Experimental configuration

DOOM-SED consists of $B = 4$ audio-text fusion blocks, each having a feature dimension of 768 and a feedforward dimension of 3,072. The model used 12 attention heads. Log-mel spectrograms were obtained with 128 mel bins, a 64-ms window, and an 8-ms hop length. Text labels were lower-cased, normalized by replacing underscores with spaces, and prefixed with “Sound of” to form text prompts. We used the text encoder of a CLAP model [10] to obtain label embeddings robust to wording variations. On the other hand, we used an audio encoder that combines a 12-layer convolutional neural network and BEATs [30] to obtain frame-level features at a temporal resolution of 64 ms. The event activities were post-processed using a median filter with a window size of 7.

Table 1: Experimental comparison of DOOM-SED and existing models. AS_{st}, Down., FSD50K, and Private. denote AudioSet Strong, training data including DESED and URBAN-SED, simulated training data of FineLAP, and private data, respectively.

ID	Model	Training data	DESED				URBAN-SED				MAESTRO Real		
			PSDS1	PSDS2	Eb-F1	AUROC	PSDS1	PSDS2	Eb-F1	AUROC	maAUC	mpAUC	Sb-F1
B1	FLAM [11]	AS _{st} & Down.	0.021	0.197	0.093	0.903	0.291	0.501	0.164	0.927	0.902	0.524	0.476
B2	FineLAP [12]	AS _{st} & FSD50K & Down.	0.324	0.534	0.482	0.971	0.418	0.554	0.425	0.945	0.865	0.456	0.442
B3	FlexSED [14]	AS _{st}	0.146	0.138	0.270	0.927	0.073	0.122	0.129	0.858	0.766	0.354	0.383
B4	DASM [15]	AS _{st}	0.470	0.606	0.309	0.975	0.066	0.097	0.030	0.831	0.887	0.554	0.504
B5	PEA-Frame [13]	Private. (14k hours)	0.354	0.440	0.424	0.975	0.132	0.165	0.164	0.896	0.838	0.434	0.446
P1	DOOM-SED	AS _{st}	0.524	0.777	0.629	0.984	0.151	0.263	0.201	0.860	0.889	0.550	0.506
P2	w/ simulated data	AS _{st} & FSD50K	0.537	0.769	0.608	0.983	0.203	0.336	0.248	0.903	0.887	0.544	0.482

Table 2: Experimental comparison of the downstream-oriented architecture and training scheme of DOOM-SED.

ID	Model	DESED				URBAN-SED				MAESTRO Real		
		PSDS1	PSDS2	Eb-F1	AUROC	PSDS1	PSDS2	Eb-F1	AUROC	maAUC	mpAUC	Sb-F1
P1	DOOM-SED	0.524	0.777	0.629	0.984	0.151	0.263	0.201	0.860	0.889	0.550	0.506
Q1	P1 w/o label-to-audio cross-attention (Eq. (2))	0.421	0.592	0.489	0.967	0.130	0.207	0.100	0.870	0.892	0.511	0.482
Q2	P1 w/o audio-to-label cross-attention (Eq. (3))	0.478	0.713	0.438	0.976	0.123	0.225	0.102	0.862	0.867	0.435	0.436
Q3	P1 w/o label self-attention (Eq. (5))	0.502	0.762	0.577	0.982	0.137	0.225	0.143	0.866	0.892	0.503	0.476
R1	P1 w/o ontology-based label resolution (Alg. 1(a)–(b))	0.501	0.725	0.588	0.979	0.141	0.264	0.195	0.852	0.892	0.505	0.486
R2	w/o ontology-based label replacement (Alg. 1(c))	0.431	0.616	0.531	0.979	0.130	0.216	0.176	0.846	0.865	0.493	0.474
R3	w/ all event labels as training input [15]	0.474	0.775	0.134	0.982	0.085	0.258	0.012	0.836	0.893	0.497	0.455

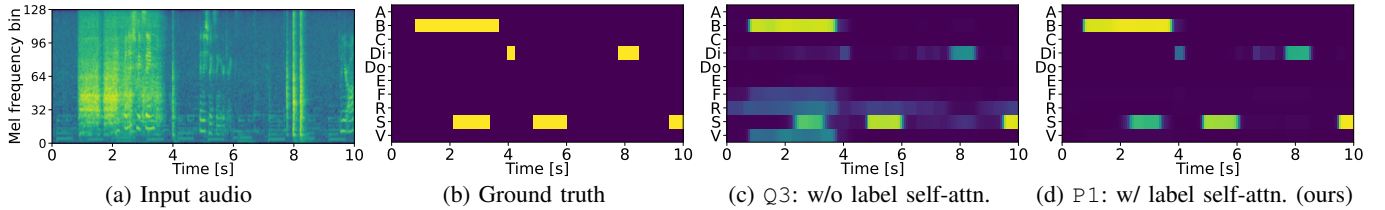


Fig. 3: Excerpts of detection results for DESED. The labels of A, B, C, Di, Do, E, F, R, S, and V represent Alarm_bell_ringing, Blender, Cat, Dishes, Dog, Electric_shaver_toothbrush, Frying, Running_water, Speech, and Vacuum_cleaner, respectively.

Our model was trained using the AdamW optimizer [31] with a learning rate of 1.0×10^{-4} and a weight decay of 1.0×10^{-4} . The number of labels used during training K was set to 10. Training was conducted for 20 epochs with a batch size of 128. An exponential moving average of the model weights was maintained during training to obtain a more stable final model. These hyperparameters were determined experimentally using the validation sets.

Performance was evaluated separately for each dataset. For DESED and URBAN-SED, we report the polyphonic sound detection scores (PSDS1 and 2) [32], [33], event-based F1 score (Eb-F1) [34], and area under the receiver operating characteristic curve (AUROC) [11]. For MAESTRO Real, we report mean AUROC (maAUC) [29], mean partial AUROC (mpAUC) [29], and segment-based F1 score (Sb-F1) [35]. These metrics follow standard protocols for each dataset.

4.3. Experimental results

The detection performance of the proposed and existing open-vocabulary SED models is summarized in Table 1. Among the existing models, the best-performing model differed across datasets. In contrast, our model (P1) is on par with or outperformed all existing methods that were not trained on the downstream datasets (B3–B5) across all datasets. In addition, on DESED, P1 outperformed the models trained on the

downstream dataset (B1 and B2). Although using the simulated data (P2) caused slight degradation on the other datasets, it significantly improved performance on URBAN-SED.

Table 2 summarizes the effects of the training strategy and network architecture on detection performance. Both text self-attention and audio–text cross-attention improved detection performance (Q1–Q3). As shown in Fig. 3, removing text self-attention (Q3) caused crosstalk for “Vacuum_cleaner,” “Blender,” and “Frying,” whereas P1 suppressed these false activations. Selecting positive and negative labels based on the ontology structure improved detection performance (R1→P1). In addition, replacing the positive labels with their ancestors also improved the performance (R2→R1). Training with all AudioSet labels as input (R3) reduced the event-based F1 score, highlighting the importance of downstream-oriented training.

5. CONCLUSION

This paper presented a training scheme and network architecture for a downstream-oriented open-vocabulary model for sound event detection. The experimental results demonstrated that the proposed DOOM-SED significantly improved the detection performance compared to existing open-vocabulary models on multiple benchmarks. Our future work includes further improving DOOM-SED to achieve performance comparable to models trained specifically for individual downstream tasks.

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